Requirements Specification

3-phase pump protection monitor (industrial-motor-monitor)

Item	Detail
Document	Requirements Specification, Rev F
Revision	F, 27 October 2026. Adds the 27 October retest, measured bias and rail voltages, relay markings and the ESP32 core version. Earlier: Rev E to Rev A (7 October 2026).
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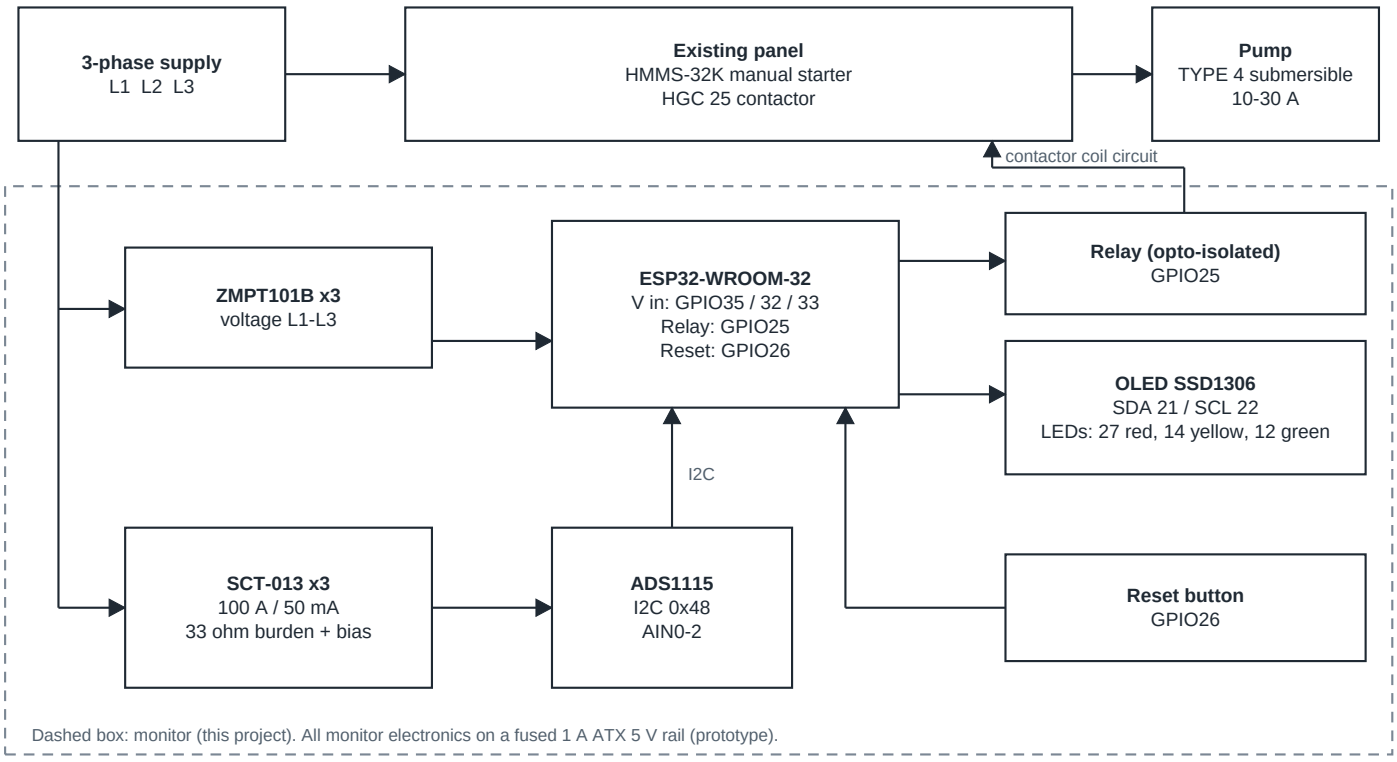
Basis	firmware/main.prototype/main.prototype.ino as committed in 875d7ea, README.md and docs/ of this repository. Line numbers (L) refer to that sketch.
Scope of this document	Describes the prototype as published here, under the permission stated in the README. The company's commercial design and proprietary material are outside this document.
Evidence statement	Bench observations are in docs/assembly-notes.md: a retest on 27 October 2026 with the committed sketch (display readings, clamp meter read in one state) and earlier observations from an earlier build. Bias and rail voltages are in docs/hardware.md. They are reported observations, not a calibration. No trip test, trip time or fail-safe test is recorded. 'Not documented' means no evidence is published here, not that the work was not done. This document makes no claim about results outside the repository. Not a certified protection device.

1. Scope

Item	Detail
System	ESP32-based monitor of three phase voltages and three phase currents that opens the contactor coil circuit on a fault.
Target	3-phase submersible drainage pumps, TYPE 4 series, 10-30 A, on a panel with a Hyundai HGC 25 contactor (coil 220-240 V AC) and an HMMS-32K manual motor starter (24-32 A).
Mains frequency	50 Hz assumed in timing analysis.
Out of scope	Replacing the HMMS-32K or HGC 25 protection. Product certification. Remote or network functions.

2. System block diagram



3. Conventions

Term	Meaning
Verif.	T = test, I = inspection, A = analysis, D = demonstration.
Implemented in code	Present in the committed sketch. Correct operation is not shown by any record in this repository.
Partly	Present in code with a limitation listed in the notes and in Section 7.
Not implemented	Absent from the committed sketch.
Document only	A requirement on documentation or build, not on code.

4. Committed parameters

Parameter	Value	Source	Note
Pump full-load current	PUMP_FLA = 10 A	L28	Set per pump in the source and re-flash.
Over-current limit	150 % of PUMP_FLA (15 A at 10 A)	L29	Basis for 150 % is not documented.
Over-current confirmation	500 ms	L33	Checked once per pass.
Voltage window	150 V to 280 V	L34-35	On a scale that is still marked 'calibrate later'.
Phase-loss level	80 V	L31	A phase below it while another is above.
Imbalance limit	25 % from the average current	L30	Only above 1 A average and after startup.
Motor-start detection	any phase above 0.5 A after being off	L44, L192-199	Off means below 0.15 A.
Startup window	5000 ms from detected start	L46, L192-204	Inhibits over-current and imbalance. Voltage stays active.
Voltage scale	100	L43	Marked 'calibrate later'.
Current chain	burden 33 ohm, CT ratio 2000, cal 1.0, noise floor 0.15 A	L38-42	CT is 100 A / 50 mA.
ADS1115	GAIN_ONE (+/-4.096 V), 860 SPS, address 0x48	L109-118	Three channels shared.
Samples per pass	60 per current channel, 200 per voltage channel	L120-124, L210	Voltage samples 400 us apart.
Filters	median of the last 3 passes, current and voltage	L170-179, L227-238	
Watchdog	8 s, panic reboot, armed after init	L127-128	Core version dependent, see R-04.

5. Derived figures (analysis from the source, not measurements)

Figure	Value	Basis
Burden voltage at the limit	0.25 V RMS at 15 A, 0.74 V RMS (1.05 V peak) at 45 A	I / 2000 x 33 ohm. Fits the +/-4.096 V range. Bias measured at 1.56 V (reported, hardware.md).
Clipping point of the current inputs	about 67 A RMS	1.56 V / 33 ohm x 2000 = 94.5 A peak. The lower swing limit governs. Above any trip limit up to 45 A.
Current resolution	about 7.6 mA per ADC count	125 uV per count / 33 ohm x 2000.
Voltage at zero input	about 0 V	The bias is now removed (L219-224), so a dead phase reads low.
Voltage sample window	about 80 ms plus read time	200 samples at 400 us. About four 50 Hz cycles.
Current sampling per channel	at most 287 SPS, under 5.7 samples per cycle	860 SPS shared by three channels. Blocking reads make the real rate lower.
Pass time	at least about 0.3 s	At least 180 current reads at 860 SPS plus the voltage window.
Over-current trip time	roughly 1.2 s or more	Two passes for the median to follow, then 500 ms confirmation checked once per pass.
Voltage trip time	roughly 0.6 s or more	Two passes for the median to follow. Not measured.

6. Requirements

6.1 Functional

ID	Requirement	Verif.	Status	Notes
REQ-F-01	The system shall measure the RMS voltage of each phase with the DC bias removed.	T	Partly	L206-238. Bias removed, 3-pass median. Matched a multimeter at 235 V, one point only. Scale still marked provisional (L43). See R-02.
REQ-F-02	The system shall measure the RMS current of each phase with the bias removed.	T	Implemented in code	L148-180. 27 Oct retest: running 3.6 A, cutting 7.1 A. Clamp meter read only with the saw plugged in and off (0 A). An earlier build agreed with a clamp meter at about 3.5 and 7.5 A (reported). See R-05, R-11.
REQ-F-03	The system shall trip when any phase current stays above 150 % of the pump's full-load current for the confirmation time, outside the startup window.	T	Implemented in code	L245-259. FLA set in the source. See R-01.
REQ-F-04	The system shall trip when any phase voltage is outside the voltage window.	T	Implemented in code	L261-266. Two passes needed through the median. Active during startup. See R-02, R-07.
REQ-F-05	The over-current and imbalance checks shall be inhibited for the startup window that opens when a motor start is detected.	T	Implemented in code	L192-204. Also opens if the monitor boots with the motor running. See R-03.
REQ-F-06	The system shall trip on phase loss and on current imbalance above the limit.	T	Implemented in code	L268-288. Thresholds 80 V and 25 % not validated. See R-10.
REQ-F-07	On a trip the system shall drive GPIO25 low to open the contactor coil circuit.	T	Implemented in code	L291-298.
REQ-F-08	A trip shall latch. Only the reset button (GPIO26) shall clear it.	T	Partly	L318-330. The latch is in RAM, so a power cycle or watchdog reboot clears it. See R-04.
REQ-F-09	The OLED shall show the state, three currents, three voltages and the limit.	I	Implemented in code	L342-381.
REQ-F-10	Green shall show not tripped, red tripped, yellow over-current present.	I	Implemented in code	L295-297, L311-315.
REQ-F-11	The display and the serial log shall show the cause of a trip.	I	Implemented in code	OLED L348-355, serial L299-308.
REQ-F-12	Trip events shall be logged with time and cause.	I	Partly	Serial log with millisecond timestamps (L80-82, L333-340). Not stored. See R-08.
REQ-F-13	Limits and calibration constants shall be named constants at the top of the source.	I	Implemented in code	L27-47. Compile-time. See R-09.

6.2 Hardware

ID	Requirement	Verif.	Status	Notes
REQ-H-01	The controller shall be an ESP32-WROOM-32.	I	Implemented in code	Pin map L14-21.
REQ-H-02	Three ZMPT101B voltage sensors shall connect to GPIO35, GPIO32, GPIO33.	I	Implemented in code	L14-16, L212-214.
REQ-H-03	Three SCT-013 100 A / 50 mA sensors shall connect through a 33 ohm burden and bias divider to ADS1115 AIN0-2 at 0x48.	I	Implemented in code	Bias divider is two 10 kohm resistors. Measured bias 1.56 / 1.56 / 1.55 V at AIN0-2 (reported, 27 Oct).
REQ-H-04	The current channels shall not clip up to and beyond the trip limit.	A	Analysis only	See Section 5. With the measured 1.56 V bias the inputs clip at about 67 A RMS.
REQ-H-05	The trip output shall be an opto-isolated relay on GPIO25 with COM-NO in series with the K1 coil.	I	Implemented in code	TONGLING JQC-3FF-S-Z, 5 V coil, contacts marked 10 A 250 V AC (reported). Pin state during reset not recorded.
REQ-H-06	The monitor shall be powered from a fused 5 V rail.	I	Implemented in code	Prototype: 1 A fuse on an ATX 5 V rail.

ID	Requirement	Verif.	Status	Notes
REQ-H-07	SSD1306 OLED on I2C (SDA 21, SCL 22, 0x3C). LEDs on GPIO27 red, GPIO14 yellow, GPIO12 green.	I	Implemented in code	L19-21, L101-104.
REQ-H-08	The monitor shall act only on the coil circuit. The HMMS-32K and HGC 25 protection shall stay connected.	I	Document only	Stated in README and safety.md.
REQ-H-09	Sensor inputs shall be protected against overvoltage and transients.	A	Not implemented	hardware.md: no external protection on the ZMPT101B outputs. See R-06.
REQ-H-10	The relay shall stay open until initialisation succeeds. An initialisation failure shall leave it open. A watchdog shall be armed.	T	Implemented in code	L95-130. Not tested in any published record. See R-04.
REQ-H-11	For installation the build shall be insulated and enclosed.	I	Document only	See safety.md.

6.3 Non-functional

ID	Requirement	Verif.	Status	Notes
REQ-N-01	Thresholds and calibration constants shall be documented with their basis.	I	Document only	hardware.md records the values. Library versions were not recorded. No basis given for 150 % FLA.
REQ-N-02	The time from fault to relay open shall be specified and measured.	T	Not documented	Target not set. See R-05.
REQ-N-03	Each requirement shall trace to a verification method and a recorded result.	I	Not documented	Bench observations are in assembly-notes.md. There is no per-requirement test log.
REQ-N-04	The firmware shall build from one source layout.	D	Partly	Real source is the Arduino sketch. Built on ESP32 core 2.0.17 (reported). Library versions not recorded.
REQ-N-05	Documentation shall state the status and that the device is not certified.	I	Document only	README and safety.md.

7. Risk register

ID	Risk	Effect	Action	Related
R-01	The limit is 150 % of one compile-time FLA value (default 10 A). The basis for 150 % is not documented. At FLA 30 A the limit is 45 A.	A wrong FLA constant gives wrong protection. A 150 % limit is far above the HMMS-32K range for large pumps.	Document the basis. Set FLA from the nameplate per pump. Keep the HMMS-32K as primary protection.	F-03, F-13
R-02	The voltage scale is still marked 'calibrate later' and was checked at one point only (235 V). The 150 V, 280 V and 80 V levels depend on it. ESP32 ADC non-linearity. Phase reference unconfirmed (about 230 V to neutral, about 400 V phase to phase). The ZMPT101B modules are powered from 3.3 V (reported).	Accuracy across the range is unknown. Trips may be wrong or missed.	Calibrate across the range and record it. Confirm what each ZMPT101B measures.	F-01, F-04, F-06
R-03	The window opens on detected motor start. It also opens if the monitor boots with the motor running. Detection lags by one pass.	Over-current and imbalance are inhibited for 5 s after a reboot with the motor running. Residual risk is low.	Record start detection and inrush in the test log.	F-05
R-04	The trip latch is in RAM. A power cycle or watchdog reboot clears it. After a reset with a fault still present, the relay is enabled until the next pass trips again. The watchdog call is core-version dependent.	A reboot re-arms the relay, so the contactor can be started again. A build on another core may not compile.	Document the behaviour. State the core version. Test reset and watchdog.	F-08, H-10
R-05	Sampling rate and latency are not measured. Blocking reads, median of 3 passes.	Poor RMS accuracy and a trip time of roughly 1.2 s or more.	Measure the rate and the end-to-end trip time.	F-02, N-02
R-06	No input protection documented on sensor inputs.	Transients can damage inputs.	Document or add clamping and series resistance.	H-09
R-07	Voltage and imbalance trips have no explicit confirmation time. Only the median gives two-pass persistence.	A disturbance lasting two passes can trip. A motor start can sag the supply.	Add a defined confirmation time. Test a start.	F-04, F-06
R-08	The event log is serial only and is not stored.	Faults cannot be reviewed without a connected host.	Store events or document the serial capture method.	F-12
R-09	Limits are compile-time constants.	A pump change needs a re-flash.	Move to configurable settings or document the procedure.	F-13
R-10	Phase-loss (80 V) and imbalance (25 %) levels are not validated. Where several faults match, the last one tested sets the reported cause. Imbalance runs only while voltage monitoring is enabled.	Wrong cause shown. Levels may be too tight or too loose.	Validate the levels. Report the first or most severe cause.	F-06, F-11
R-11	With the saw plugged in and switched off, the monitor reads about 1.5 A on all three phases while a clamp meter reads 0 A (27 Oct retest). With the saw unplugged it reads about 0.1 A. The cause is unresolved. The sketch treats the motor as off only below 0.15 A.	A measurement effect that overstates small currents. While it is present the start-detection flag does not reset, so the startup window may not reopen on a later start and the inrush could trip the monitor.	Find the cause in the CT and jack path or in the surroundings. Retest with the effect removed. Do not hide it with a fixed offset.	F-02, F-05

No requirement has a per-requirement test record. A test log, a calibration across the range and a measured trip time would move requirements from Implemented in code to verified.